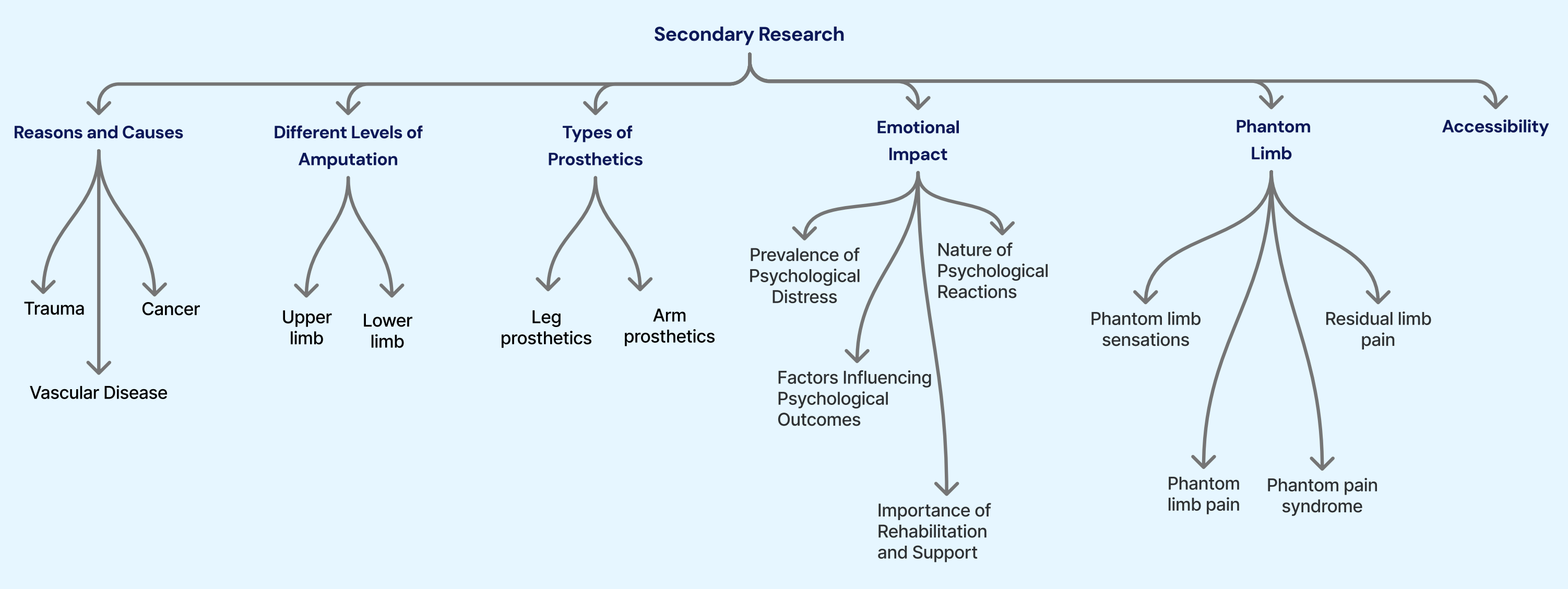
Design For Special Needs

Introduction

Amputees are people who have gone through the amputation surgery and have lost a limb. There are many mental and emotional changes that they go through and it can affect basic mobility, independence, career prospects, and even self-identity.

There are many people living their lives after having their limbs amputed, our aim is not only to understand the physical, mental and emotional changes that these amputees go through but also get a wholistic view of the impact this has on the people around them and how they deal and process this happening in their lives.

Secondary research



Secondary Research

- Reasons and Causes
  - Trauma
  - Cancer
  - Vascular Disease
- Different Levels of Amputation
  - Upper limb
  - Lower limb
- Types of Prosthetics
  - Leg prosthetics
  - Arm prosthetics
- Emotional Impact
  - Prevalence of Psychological Distress
  - Nature of Psychological Reactions
  - Factors Influencing Psychological Outcomes
  - Importance of Rehabilitation and Support
- Phantom Limb
  - Phantom limb sensations
  - Residual limb pain
  - Phantom limb pain
  - Phantom pain syndrome
- Accessibility

Interview Insights

1. our main insight was that amputees had major concerns over **affordability**, **comfort** and budget of the prosthetics

2. our main insight was **multiple factors go into consideration** while selecting the prosthetic and prioritising the right factors can help prevent buying the wrong match of prosthetics

3. our main insight was **people's beliefs and support systems play a major role** in their emotional and mental journey after such incidents

4. our main insight was while **modern tools help the amputees be more independent** and manage their daily lives easily, it is more important to maintain good emotional support around oneself

5. our main insight was **access to a variety** of options of prosthetics and assistive devices can help make one's life easier

Problem Areas

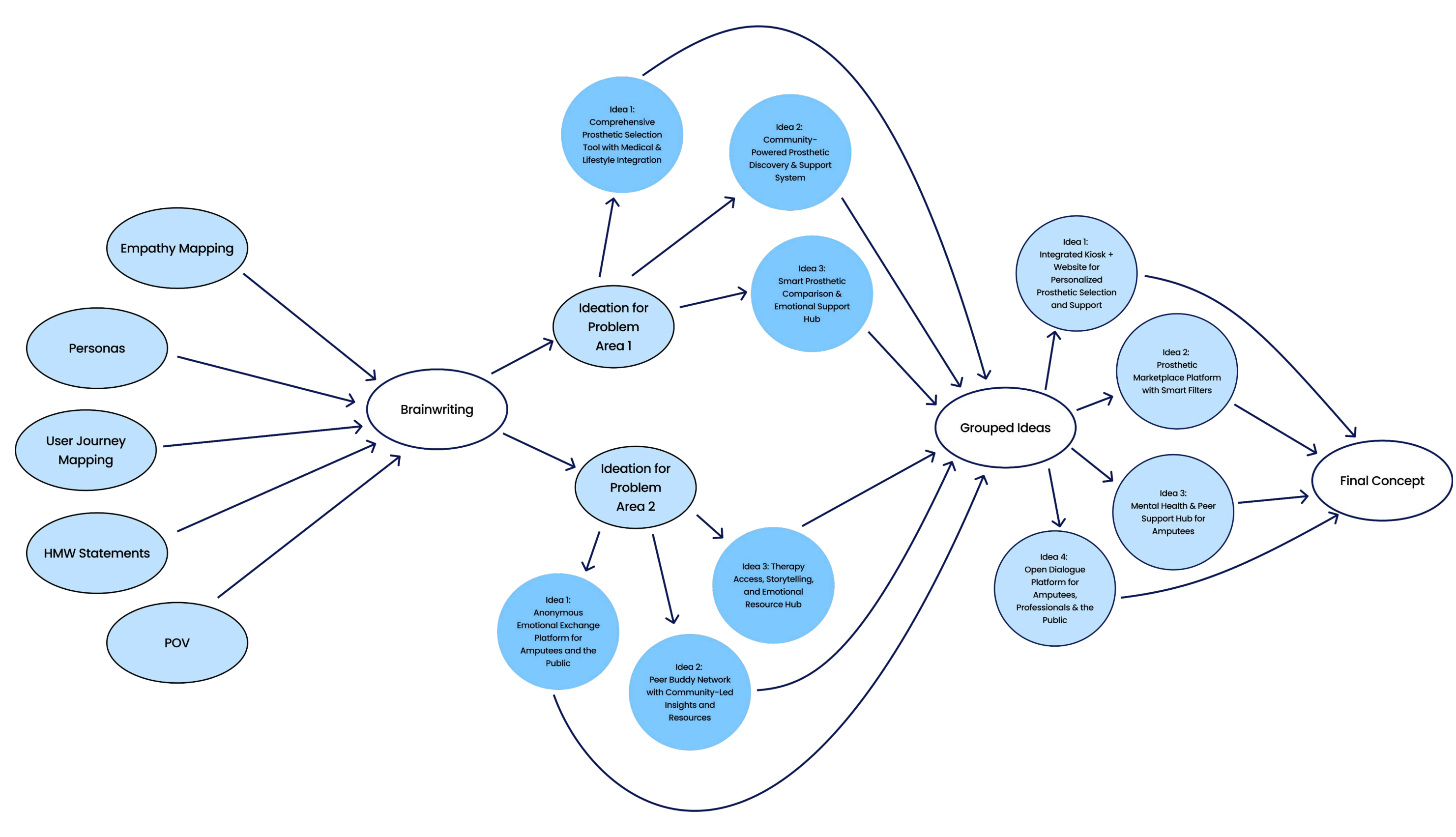
Problem Area 1

After the amputation, choosing the right prosthetic is very important. With growing technology ands access there is an overwhelming number of options available to the patients, with no clear guidance from one reliable source on which to select. This leads to people making huge medical and personal decisions while still adjusting to life post-amputation, with little support and lots of pressure.

Problem Area 2

Along with the physical recovery, the emotional and psychological recovery is extremely important as well. Which often gets overlooked due to various reasons. Apart from the mandatory therapy patients don't have the comfort and safe space to speak freely about these aspects.

Ideation Mapping



Empathy Mapping

Personas

User Journey Mapping

HMW Statements

POV

Brainwriting

Ideation for Problem Area 1

Idea 1: Comprehensive Prosthetic Selection Tool with Medical & Lifestyle Integration

Idea 2: Community-Powered Prosthetic Discovery & Support System

Idea 3: Smart Prosthetic Comparison & Emotional Support Hub

Ideation for Problem Area 2

Idea 1: Anonymous Emotional Exchange Platform for Amputees and the Public

Idea 2: Peer Buddy Network with Community-Led Insights and Resources

Idea 3: Therapy Access, Storytelling, and Emotional Resource Hub

Grouped Ideas

Idea 1: Integrated Kiosk + Website for Personalized Prosthetic Selection and Support

Idea 2: Prosthetic Marketplace Platform with Smart Filters

Idea 3: Mental Health & Peer Support Hub for Amputees

Idea 4: Open Dialogue Platform for Amputees, Professionals & the Public

Final Concept

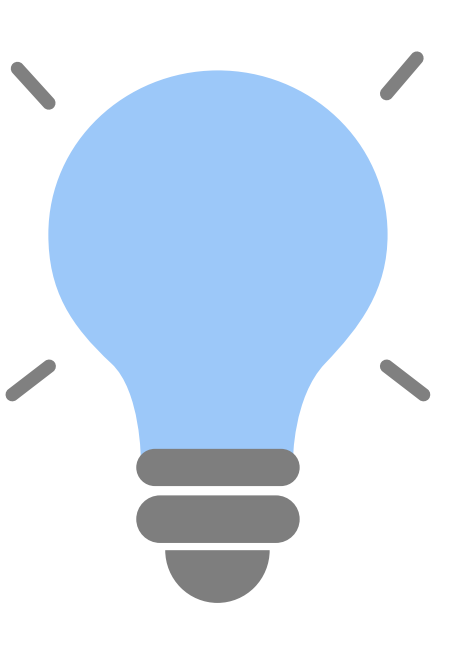
Grouped Ideas – 4

Idea 1

The first idea was a **kiosk+website** digital product which focuses on the user (amputees) finding the right prosthetic for them by streamlining and grouping all the factors taken into consideration while selecting the same.

Idea 2

The second idea gives the user a **platform** where they can browse through all the prosthetics available in the market and filter according to what they want



Idea 3

The third idea suggests a website+app which will be the one stop destination for any mental health support that the user may need, with **community support** and guidance to get the right help

Idea 4

The fourth idea focused on the **interaction** between amputees, medical professionals and non-amputees in a casual way and focusing on mental health and being more open about the same.

Final Problem statement

After an amputation, patients **struggle to find** the right prosthetic suited to their **budget, lifestyle, and medical needs.** With **scattered, technical information** and little personalised guidance, they're often forced to make **overwhelming decisions** without clarity or confidence.

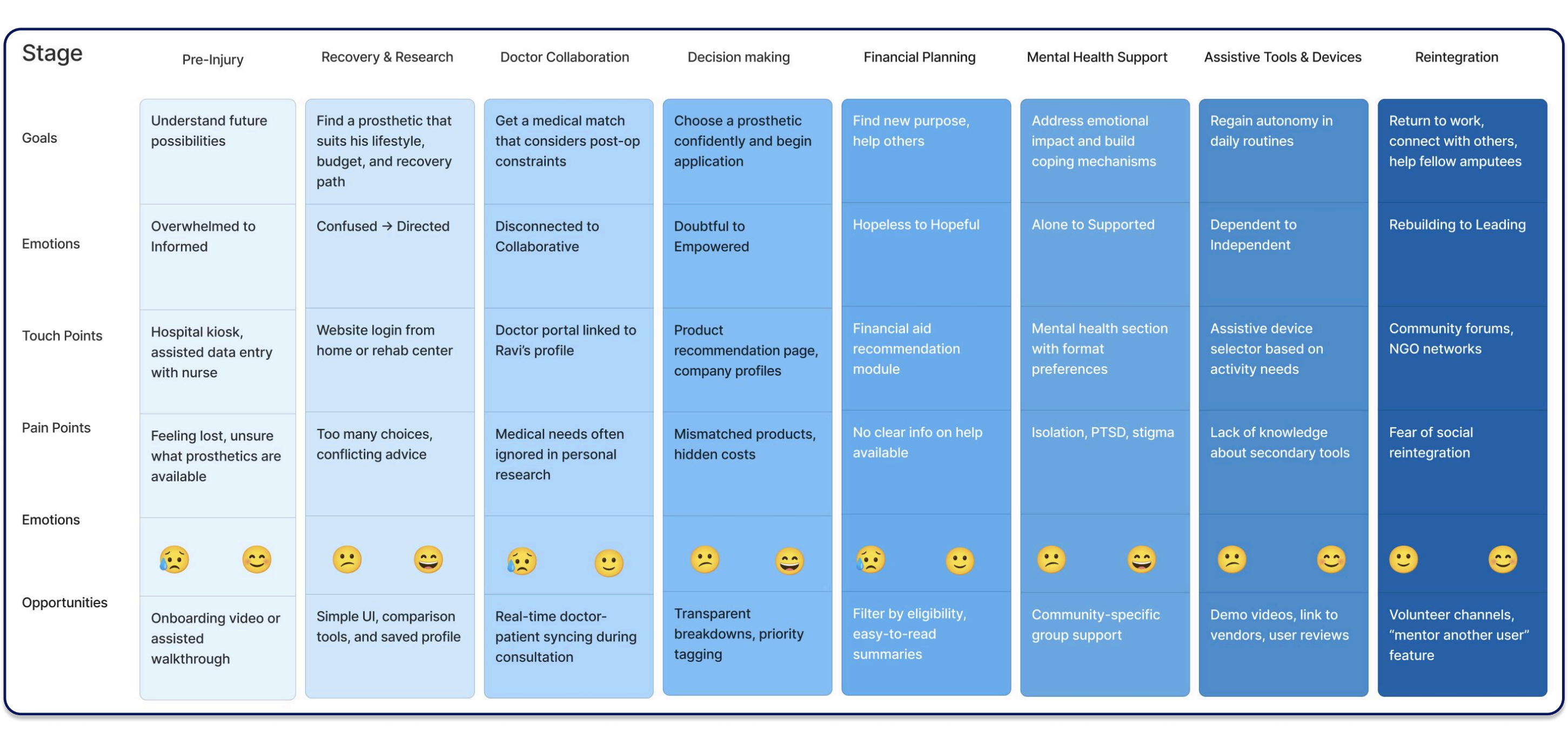
Final Concept

What system/platform are we proposing?
We're proposing AmpUnity, a digital platform available primarily through an in-hospital kiosk and additionally as a website. The kiosk allows amputees to enter lifestyle, medical conditions and financial details while they wait, with staff assistance if needed. Doctors can access this and give input during consultations using the patient's unique ID. The web version acts as a secondary access for patients to revisit and update inputs from home. AmpUnity offers tailored prosthetic recommendations, daily support tools, and mental health resources.

How does it work?
Patients receive a unique ID to log into the kiosk or website. They enter details like career, mobility requirements, hobbies, type of amputation and budget. Doctors input medical constraints. The system then generates a ranked list of prosthetics with tags like "sport-approved" or "budget-friendly," including details like pricing, delivery timeline and maintenance. It also suggests funding options and supportive tools, along with access to tools and mental health resources, aiming to holistically support amputees through recovery and reintegration.

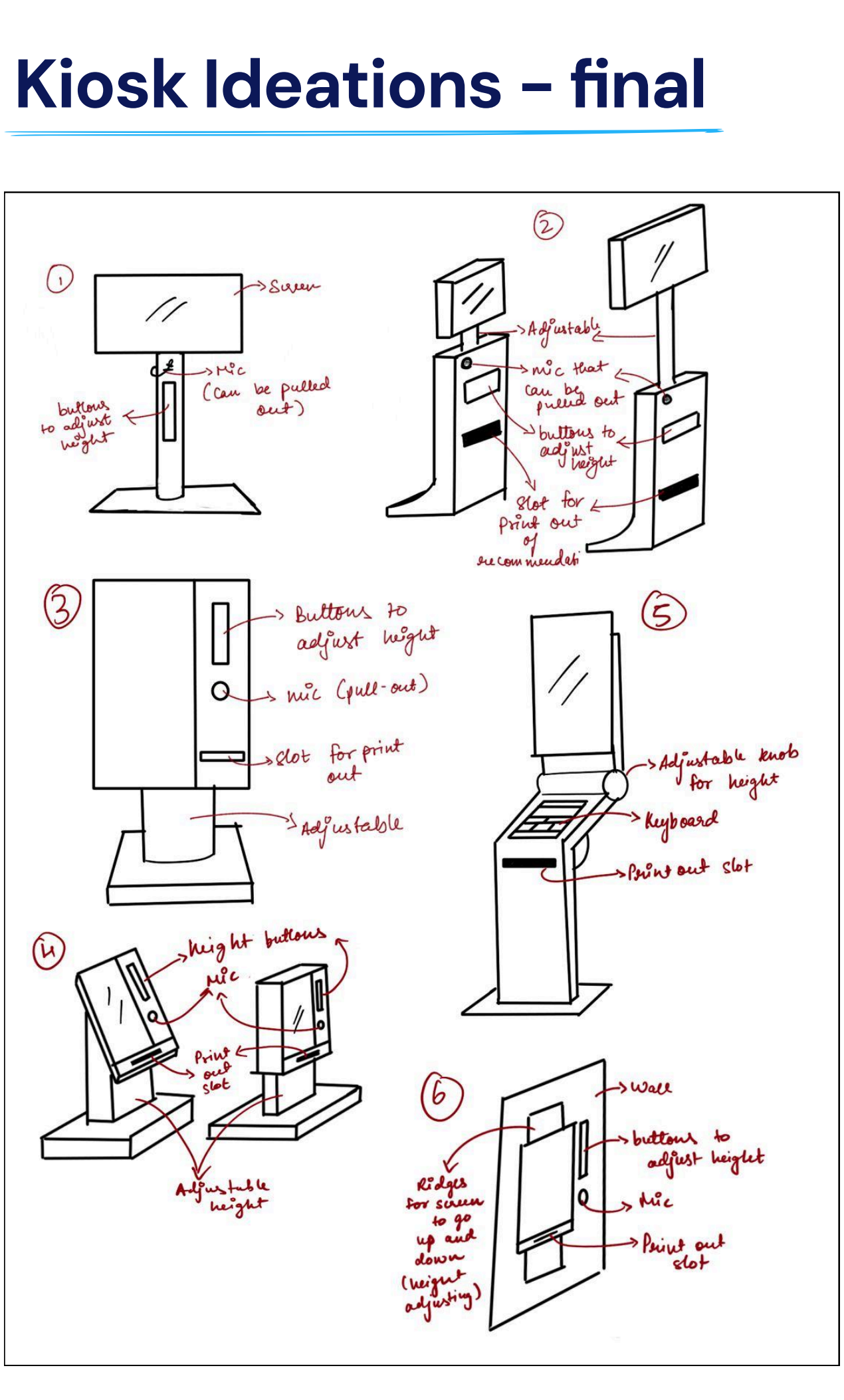
How does it help the current problem at hand?
AmpUnity simplifies a complex and overwhelming decision by combining personal, lifestyle, and medical inputs to provide clear, patient-specific prosthetic recommendations. It avoids the confusion of generic product ratings, supports decision-making with doctor-backed suggestions, and includes funding and mental health support. The platform, combining kiosk and website, bridges gaps for those with limited tech access, informational gaps and supports more confident, informed, and timely decisions.

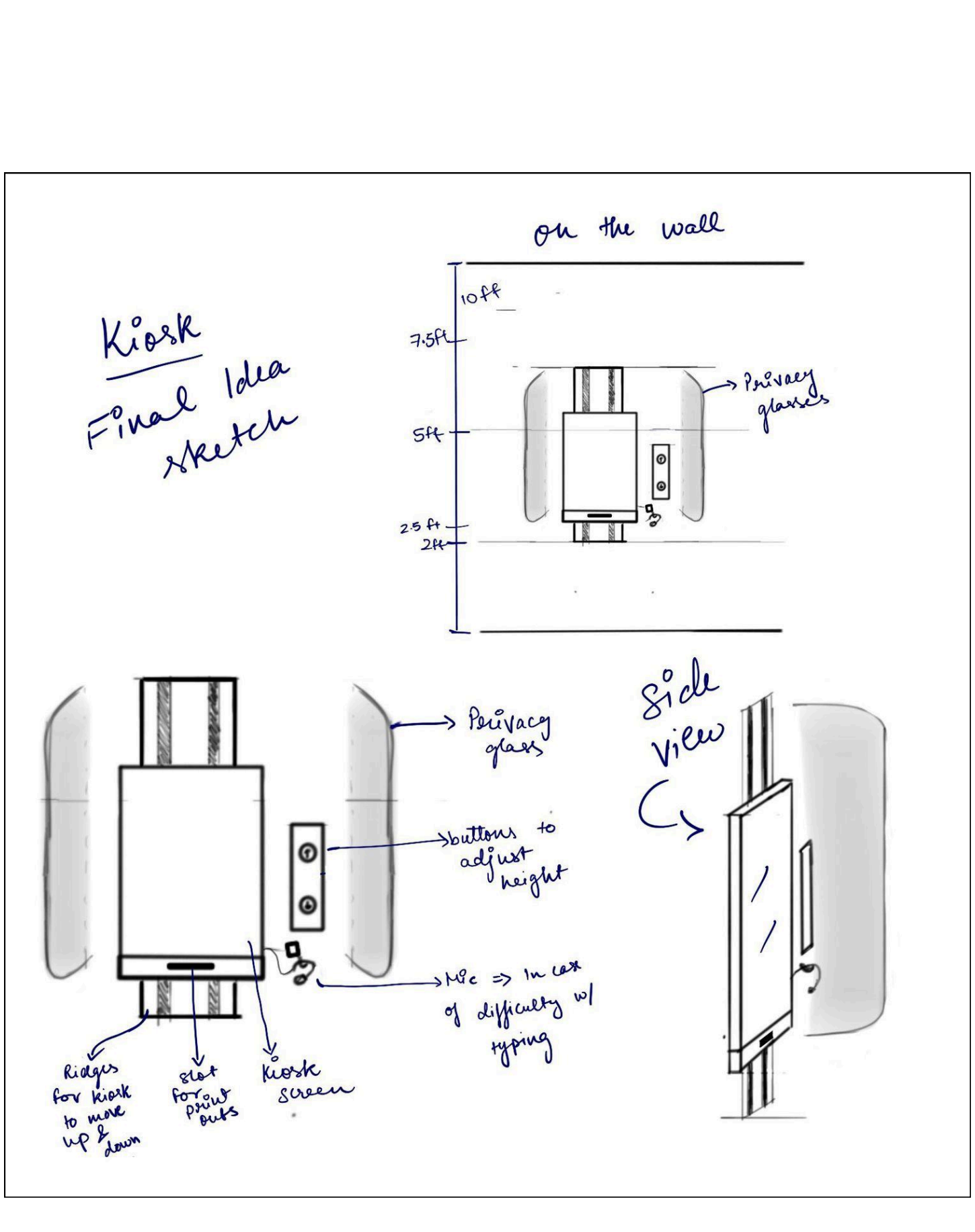
User Journey Map



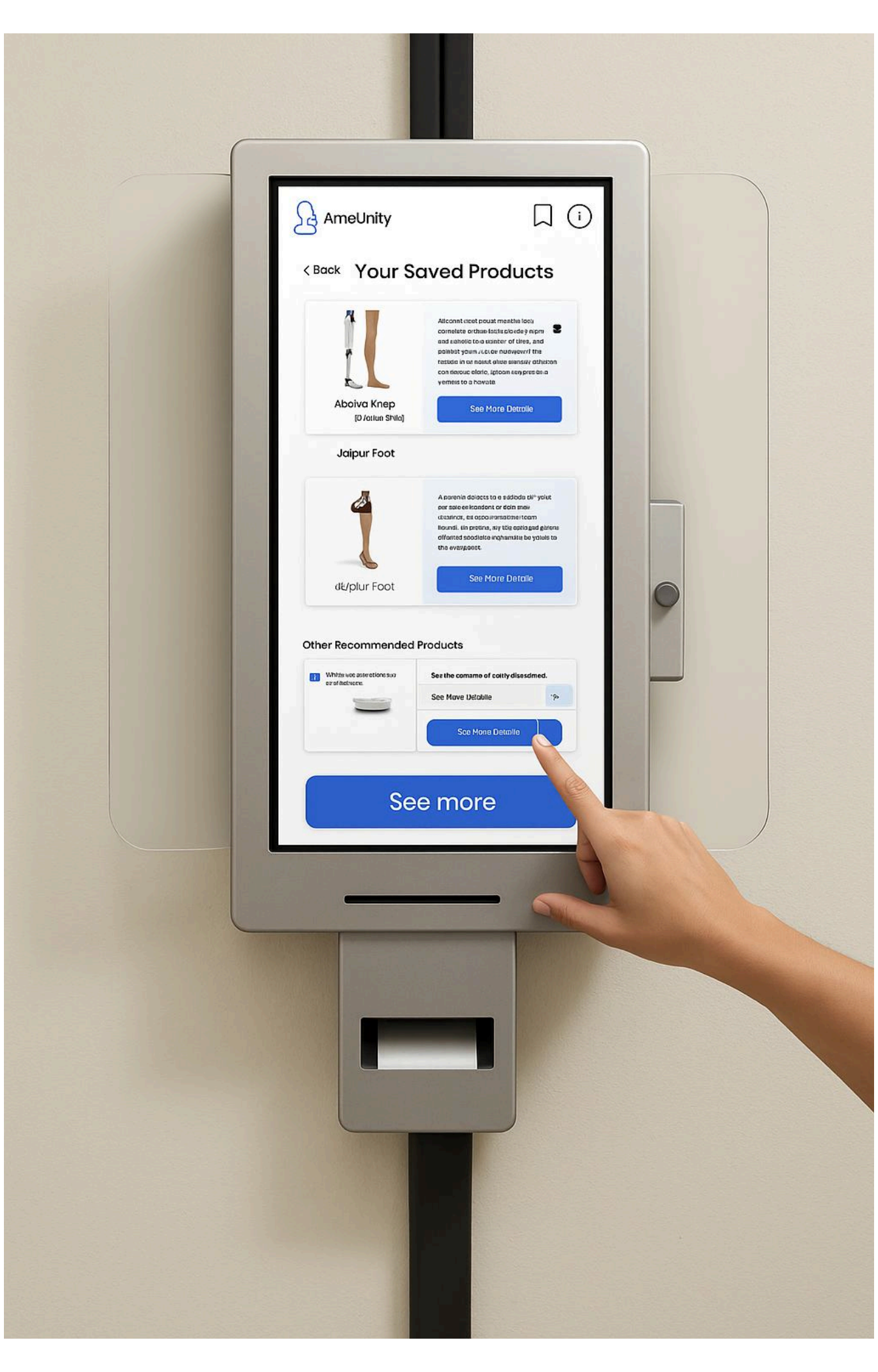
Stage	Pre-Injury	Recovery & Research	Doctor Collaboration	Decision making	Financial Planning	Mental Health Support	Assistive Tools & Devices	Reintegration
Goals	Understand future possibilities	Find a prosthetic that suits his lifestyle, budget, and recovery path	Get a medical match that considers post-op constraints	Choose a prosthetic confidently and begin application	Find new purpose, help others	Address emotional impact and build coping mechanisms	Regain autonomy in daily routines	Return to work, connect with others, help fellow amputees
Emotions	Overwhelmed to Informed	Confused → Directed	Disconnected to Collaborative	Doubtful to Empowered	Hopeless to Hopeful	Alone to Supported	Dependent to Independent	Rebuilding to Leading
Touch Points	Hospital kiosk, assisted data entry with nurse	Website login from home or rehab center	Doctor portal linked to Ravi's profile	Product recommendation page, company profiles	Financial aid recommendation module	Mental health section with format preferences	Assistive device selector based on activity needs	Community forums, NGO networks
Pain Points	Feeling lost, unsure what prosthetics are available	Too many choices, conflicting advice	Medical needs often ignored in personal research	Mismatched products, hidden costs	No clear info on help available	Isolation, PTSD, stigma	Lack of knowledge about secondary tools	Fear of social reintegration
Emotions	😥 😊	😕 😄	😥 🙂	😕 😄	😥 🙂	😕 😄	😕 😊	🙂 😊
Opportunities	Onboarding video or assisted walkthrough	Simple UI, comparison tools, and saved profile	Real-time doctor-patient syncing during consultation	Transparent breakdowns, priority tagging	Filter by eligibility, easy-to-read summaries	Community-specific group support	Demo videos, link to vendors, user reviews	Volunteer channels, "mentor another user" feature

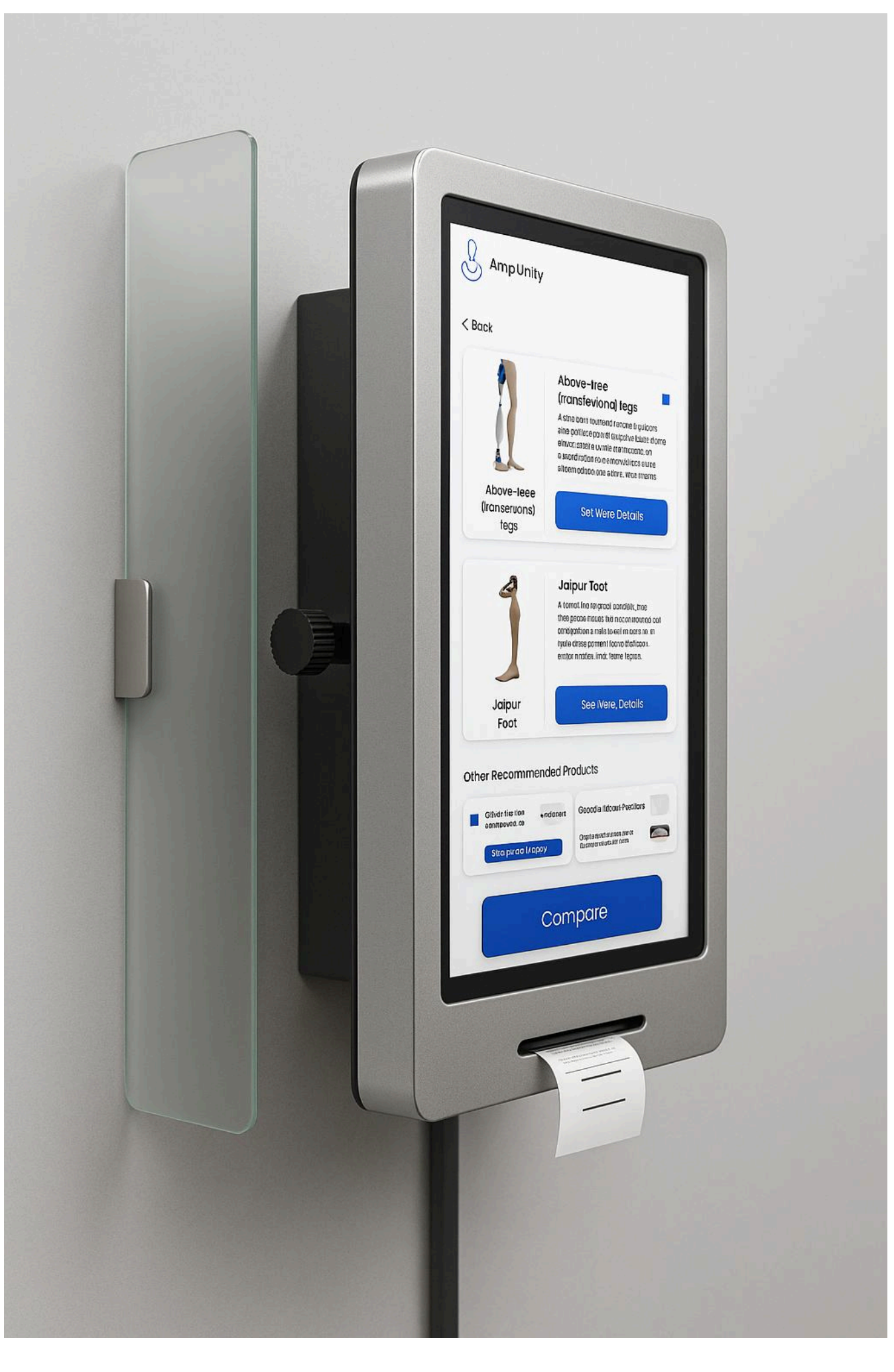
Kiosk Ideations – final





Prototype





THANK YOU